How Residual-Stream Updates Travel Through Depth

A Pre-Registered Study of Cross-Layer Dynamics in Transformers

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Abstract

Transformer components communicate through an additive residual stream, so the update a layer writes, $\Delta_\ell = v_{\ell+1} - v_\ell$, is the natural unit of cross-layer computation. We pre-registered a two-part hypothesis about these updates on factual-recall inputs — (H1) they are predictable across depth, and (H2) the prediction kernel is approximately stationary in depth — and ran the frozen protocol to completion on Pythia-410m-deduped, with replications on six models from three architecture families (Pythia, GPT-2, Qwen2). Four findings survive an adversarial battery of pre-registered nulls and post-hoc pressure tests. **(1)** H1 holds — Δ_ℓ is autoregressively predictable far above scrambled and context-perturbation nulls — but an exact variance decomposition shows the raw score (held-out $R^2 = 0.65$) is part running-sum tautology (0.52) and part genuine structure (0.13, rising to 0.44 when refitted on the persistence-stripped stream), and an untrained-network control shows the genuine part is *substantially architectural rather than learned*, on two families. **(2)** H2 fails decisively: kernels fit on one depth band predict another band's updates worse than that band's mean, so depth dynamics are pervasively non-stationary, the effective AR order is $k^* \approx 1$, and the right abstraction is a first-order, fully depth-indexed state-space model (SSM) — validating the unconstrained per-layer-pair parameterisation of cross-layer transcoders. **(3)** Fitted over frozen sparse-autoencoder sources, the SSM reconstructs Δ_ℓ above a matched per-layer dictionary at every sparsity ($+0.049$ $[+0.046, +0.052]$ R^2 at matched rank; $+0.008$ $[+0.005, +0.010]$ at matched total parameters), with the gain growing monotonically with depth within the Pythia family and replicating in sign in GPT-2 and Qwen2; roughly half of it is already present at random initialisation. The learned state is contractive at every layer (spectral radius median 0.43, memory ≈ 0.8 layers) — the mechanism behind $k^* \approx 1$ — and is *functionally informative*: a 32-dimensional state decodes the recalled answer at mid-depth better than dimension-matched PCA summaries of the very codes it is built from. **(4)** The pre-registered causal existence proof returns a clean null: no single feature's ablation moves the answer probability by the pre-registered threshold, and the group-ablation effect that does is magnitude-driven — a norm-matched *random direction* reproduces it — so factual recall is a distributed, mass-driven code with no privileged sparse subset, a conclusion that survives swapping the dictionary. As a by-product we isolate a practical failure mode: TopK dictionaries make single-feature attributions seed-unidentifiable (decoder match 0.45) where L1 dictionaries on the same activations are perfectly stable (1.00). All gates, thresholds, and fallback branches were written down before data collection; every headline carries a paired bootstrap confidence interval; and every figure regenerates from tracked tables with one script.

1 Introduction

Mechanistic interpretability decomposes transformer activations mostly layer by layer. Sparse autoencoders (SAEs) recover human-legible features from a single layer's residual stream (Cunningham et al., 2024; Bricken et al., 2023; Templeton et al., 2024), but a per-layer dictionary

answers only a per-layer question — *what features live here?* — and not the question that matters for circuit analysis and causal editing: *how does what a layer writes relate to what earlier layers wrote?* The field has begun to attack this directly. Crosscoders fit one dictionary across all layers (Lindsey et al., 2024); multi-layer SAEs (MLSAEs) find that the same latent fires at different depths on different prompts (Lawson et al., 2025); and cross-layer transcoders (CLTs), the substrate of attribution-graph circuit tracing, give every feature a *free* decoder at every downstream layer — $O(L^2)$ independent decoder matrices with no prior connecting them (Ameisen et al., 2025).

This paper asks whether that free-parameter budget is load-bearing or redundant, and it asks the question the way one asks it in signal processing: by modelling the *increments*. Because the residual stream is additive, $v_{\ell+1} = v_\ell + \Delta_\ell$, regressing states on states trivially recovers the identity; the non-trivial object is the update $\Delta_\ell = v_{\ell+1} - v_\ell$ — what the layer *adds* — exactly as differencing removes the trend in classical time-series analysis. We pre-registered two nested hypotheses about these updates on factual-recall inputs:

H1 (depth-local predictability). Δ_ℓ is predictable from the preceding updates $\Delta_{\ell-1}, \Delta_{\ell-2}, \ldots$ by a low-rank linear map, above scrambled-activation and context-perturbation nulls — residual updates form an autoregressive process along depth.

H2 (approximate stationarity). The prediction kernel is approximately shared across depth, so that a single block-Toeplitz (convolutive) kernel $\Delta_\ell \approx \sum_k W_k\, s_{\ell-k}$ over sparse per-layer sources is a useful generative model. Under H2, CLT's $O(L^2)$ decoder matrices would collapse to $O(K)$ shared kernel taps.

The hypothesis was motivated by an analogy to convolutive blind source separation in high-density surface electromyography, where sources mix into recordings through characteristic propagation kernels (Holobar and Zazula, 2007); the analogy generated the hypothesis and nothing else — the formal models and tests are native to interpretability and classical statistics. Crucially, the study was *pre-registered*: a frozen protocol fixed the diagnostics, the go/no-go gates, the null controls, the decomposition architecture as a function of the gate outcomes, and the causal-evaluation thresholds, before any data were collected. Where the data contradicted the more exciting branch, we took the pre-registered exit. Both hypotheses could have failed cheaply; what actually happened is more interesting than either clean outcome.

What we found. H1 survives every control we could construct — including two nulls specified in advance and a post-hoc pressure battery designed to break it — but decomposes on inspection: an exact (Pythagorean) split of the headline autoregressive R^2 shows that on factual prompts $0.65 = 0.52$ running-sum tautology $+ 0.13$ genuine structure, and an untrained-network control shows the genuine part is *substantially architectural* — a randomly initialised transformer exhibits it at comparable strength, on two model families (Section 3). H2 fails decisively and non-negotiably: an AR kernel fit on one third of the depth axis predicts another third's updates *worse than that band's own mean*, at every rank, with no clean segment boundary (Section 4). The pre-registered consequence is that the generative model must be a first-order, fully depth-indexed state-space model (SSM) — and that CLT's refusal to share parameters across depth is *validated*, not wasteful.

Fitting that SSM over frozen per-layer SAE sources yields the paper's central positive result (Section 6): cross-layer state reconstructs residual updates above a matched per-layer dictionary at every sparsity ($+0.049$ R^2 $[+0.046, +0.052]$ at matched rank, $+0.008$ $[+0.005, +0.010]$ at matched total parameters), the gain grows monotonically with depth (6 layers $+0.008 \rightarrow$ 24 layers $+0.050$ within Pythia; replicated in GPT-2 and Qwen2), and about half of it is present at random initialisation — so the method captures a genuinely learned signal roughly equal in size to the architectural one. The learned state-transition operator is contractive everywhere (spectral

radius median 0.43, implied memory $\approx$ 0.8 layers), which *explains* the observed first-order structure, and the state is functionally informative: at matched dimension it decodes the recalled answer at mid-depth better than PCA summaries of the codes it is built from.

The pre-registered causal test — find a feature no baseline finds, ablate it, predict the behavioural consequence — returns a null we consider one of the paper's main contributions (Section 6.6). No single feature clears the pre-registered effect threshold; ablating each example's top-100 co-firing features does clear it, but a *norm-matched random direction* reproduces that group effect, so it is a magnitude effect, not evidence of a privileged code. The null is dictionary-independent: a cleaner L1 dictionary concentrates single-feature effects five-fold and still finds no bottleneck. Factual recall in this model is a distributed, mass-driven code.

Contributions.

1. A pre-registered, gate-structured characterisation of residual-update dynamics: H1 holds (with an exact accounting of how much is tautology, and evidence the remainder is substantially architectural); H2 fails (pervasive non-stationarity, $k^* \approx 1$, query-position-specific and circuit-specific kernels); both replicate across six models, three families, and on generic Pile text (Sections 3 and 4).

2. A first-order, fully depth-indexed SSM decomposition of Δ_ℓ over sparse per-layer sources, with a reconstruction gain over matched per-layer baselines that is CI-significant, depth-graded, and approximately half learned / half architectural (Sections 5 and 6).

3. A three-axis comparison against the unconstrained $O(L^2)$ residual-write CLT parameterisation: at matched per-decoder rank the CLT keeps a small significant edge; at matched *total* parameters the SSM Pareto-dominates up to 52M parameters; below $\sim$1k training examples the SSM generalises better (Section 6.4).

4. A clean, fully controlled causal null: recall is distributed and magnitude-driven, with no privileged sparse subset — established with matched-count, matched-mass, and norm-matched random-direction controls, and robust to the dictionary (Section 6.6).

5. Two structural positives that survive every robustness check: the contractive propagation operator (gauge-invariant spectrum, seed cosine 0.999, dictionary cosine 0.993), and the functional informativeness of the SSM state for answer identity at mid-depth (Sections 6.7 and 6.8).

6. A practical warning for the field: TopK dictionaries make single-feature decoder directions seed-unidentifiable (match@0.7 = 0.45) where matched L1 dictionaries trained on the same activations are perfectly stable (1.00) — an identifiability failure that lives in the sparsity mechanism, not the model or the decomposition (Section 6.5).

Everything reproduces from tracked run tables via one script; the protocol documents were frozen before data collection and every deviation is logged (Sections A and E).

2 Related work

The residual stream and superposition. Elhage et al. (2021) formalised the framing every cross-layer method inherits: attention and MLP blocks read from and write to a shared additive channel, so features persist through depth because the channel is a running sum. Elhage et al. (2022) explains why single neurons are not the unit of analysis: models encode more features than dimensions, in overlapping directions. Together these motivate both the object of this study, the per-layer write Δ_ℓ, and its principal statistical hazard — consecutive updates mechanically share a $\mp v_\ell$ term (Section 3.1), which we spend considerable effort separating from genuine structure.

Per-layer sparse dictionaries. SAEs trained on one layer's activations recover monosemantic features (Cunningham et al., 2024; Bricken et al., 2023) and scale to frontier models (Templeton et al., 2024). TopK SAEs replace the L1 penalty with a hard sparsity constraint (Makhzani and Frey, 2013; Gao et al., 2025). We use public per-layer suites as frozen sources — the MLSAE release for Pythia (Lawson et al., 2025), the reformatted GPT-2-small residual suite (Bloom, 2024), an open Qwen2 suite (Chanin, 2024), and EleutherAI's Pythia-70m suite (EleutherAI, 2024) — and additionally train matched L1 and TopK dictionaries on our own cached activations, which turns out to matter: the sparsity mechanism, not the model, controls whether single-feature decoder directions are identifiable across seeds (Section 6.5).

Cross-layer dictionaries. Crosscoders fit one dictionary with per-layer decoders across all layers at once (Lindsey et al., 2024). MLSAEs share the dictionary and let activations vary per layer; their headline observation — a latent is often active at exactly one layer per token, but at *different* layers across tokens — already strains the picture of a feature with a fixed home depth (Lawson et al., 2025). RouteSAE adds a router that picks which layer a shared SAE decodes per token (Shi et al., 2025): a *where-to-read* mechanism, complementary to the *how-updates-propagate* question here. Jacobian SAEs sparsify the causal sensitivities across a single MLP block (Farnik et al., 2025) — causal rather than correlational cross-layer structure, but with no assumption (or test) of depth-shared form.

Cross-layer transcoders. CLTs (Ameisen et al., 2025) are the closest prior method and this paper's primary conceptual baseline. A CLT feature reads at one source layer ℓ' and writes into every layer $\ell \geq \ell'$ through a separate, fully free decoder per (source, target) pair — $O(L^2)$ decoder matrices with no parameter sharing along depth. Our H2 is precisely the hypothesis that those matrices compress to a kernel in the depth difference $\ell - \ell'$; H2's failure (Section 4) is therefore an independent *validation* of CLT's unconstrained parameterisation, and our SSM-vs-CLT comparison (Section 6.4) asks the surviving quantitative question: how much reconstruction do the free per-pair decoders buy over a first-order depth-indexed state at matched sources?

Depth-wise lenses. The Tuned Lens learns per-layer affine probes that decode intermediate residual states into the final-output distribution (Belrose et al., 2023). We use it twice: its implied per-layer update $\hat{r}_{\ell+1} - \hat{r}_\ell$ is the calibrated baseline our cheap H1 diagnostic must beat before the expensive tests run (it does, Section 3.2), and it reappears as an update-prediction baseline in the evaluation, where — like every method not trained on Δ_ℓ — it scores below zero (Section 6.3).

Factual recall. We test on subject–relation–object recall because it is the best-instrumented circuit family in the literature: causal localisation to mid-depth MLPs (Meng et al., 2022; Dai et al., 2022), a staged enrich-then-extract account (Geva et al., 2023), and a standing caveat that causal-tracing locations and editing locations dissociate (Hase et al., 2023). Our data derive from CounterFact (Meng et al., 2022), with Pararel (Elazar et al., 2021) recorded in the protocol as the scaling path we ultimately did not need (Section A). Our distributed-code null (Section 6.6) is consistent with this literature's drift away from single-component stories of recall.

Signal-processing motivation. Convolution Kernel Compensation decomposes high-density surface EMG under a convolutive mixing model (Holobar and Zazula, 2007). The analogy — sources with characteristic propagation patterns along a depth-like axis — supplied H2's form (a shared kernel in the depth difference). It supplied nothing else: every model fit here is ridge regression, LASSO, alternating minimisation, or gradient descent on a linear SSM, and H2 was treated as a hypothesis to kill, which the data did.

Infrastructure. Activations are collected with TransformerLens (Nanda and Bloom, 2022) from Pythia (Biderman et al., 2023), GPT-2, and Qwen2 checkpoints; generic text is drawn from the Pile (Gao et al., 2020). Optimisation uses FISTA (Beck and Teboulle, 2009) for the sparse-code updates; the Granger-style tests are calibrated against a Freedman–Lane permutation scheme (Freedman and Lane, 1983) in Section C.

3 Phase 1 — is the residual update predictable across depth?

3.1 Study design

Pre-registration. The full protocol — four phases, per-step procedures, three go/no-go gates, null controls, decision matrices, and the causal-evaluation thresholds — was written and frozen before any data were collected. Progress and deviations were recorded exclusively in a separate tracker so that no pre-registered bar could drift. Every deviation from the plan (all quantitative-threshold adjustments explicitly sanctioned by the plan's own tuning rules, plus one descoping decision) is enumerated in Section A. The three gates: Gate 1 (is H1 supported above null?), Gate 2 (does H2 hold? — this picks the Phase-3 architecture), Gate 3 (does the decomposition reconstruct competitively?). A failed gate routes to writing up the negative result, not to silent continuation.

Primary model and notation. The primary model is Pythia-410m-deduped (Biderman et al., 2023): $L = 24$ blocks, residual dimension $D = 1024$, parallel attention/MLP blocks, with a public Tuned Lens and a public per-layer residual SAE suite. We write $v_\ell \in \mathbb{R}^D$ for the residual state after block ℓ (v_0 = embedding output; v_L = final pre-unembedding state, so 25 states per example) and

$$\Delta_\ell \;=\; v_{\ell+1} - v_\ell, \qquad \ell \in \{0, \ldots, L-1\},$$

for the update block ℓ writes. Cross-model replications use Pythia-70m (L=6) and Pythia-160m-deduped (L=12), GPT-2 small/medium/large (L=12/24/36), and Qwen2-0.5B (L=24) — six models, three families, depths 6–36.

Dataset. From CounterFact (Meng et al., 2022)[1] we keep triples the model actually knows: top-1 prediction equals the object and top-1 probability exceeds 0.1. Pythia-410m-deduped confidently knows only ∼2,000 of the 21,601 CounterFact facts, so the plan's tuning rules were exercised in the sanctioned direction (probability threshold $0.3 \to 0.1$; per-class floor $500 \to 300$), yielding **1,555 prompts in three relation classes** (organization 711, geography 511, language 333; occupation and religion fell below the floor and were dropped). Scoping to these three classes is Limitation 3 (Section 7). For every prompt we collect the residual trajectory $v_0, \ldots, v_{24}$ *at the final prompt token* — the query position, immediately before the model emits the object — giving a $[1555, 25, 1024]$ store. A fixed split (1,244 train / 311 test, seed 0) is shared by all Phase-1–3 analyses. Sanity check: applying the model's final LayerNorm and unembedding to the collected v_L recovers the model's top-1 answer (5/5 spot checks).

The $\pm v_\ell$ tautology, stated up front. Consecutive updates mechanically overlap: $\Delta_\ell = v_{\ell+1} - v_\ell$ and $\Delta_{\ell-1} = v_\ell - v_{\ell-1}$ share the term $\mp v_\ell$. Because a few residual directions carry a large, slowly drifting share of state variance (the top direction alone is 32% of state variance on the factual store, 96% on generic text), a linear map can predict Δ_ℓ from $\Delta_{\ell-1}$ partly by reading this shared persistent component — running-sum drift, not depth computation. Every AR-on-Δ number in this paper is therefore reported twice: raw, and after the tautology accounting of Section 3.4. The pre-registered scrambled null does *not* separate the two (permuting examples

[1] Via the s–r–o-ordered mirror `NeelNanda/counterfact-tracing`.

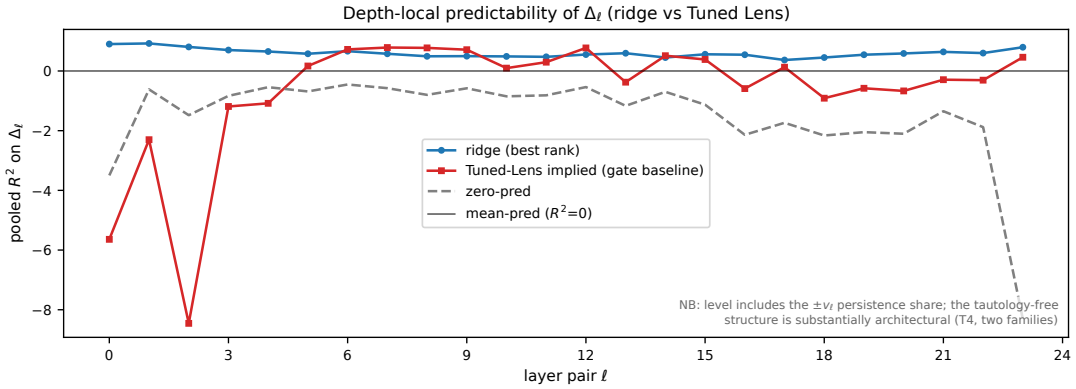


Figure 1: Per-layer-pair held-out R^2 of low-rank ridge predicting Δ_ℓ from v_ℓ on Pythia-410m-deduped, against the mean, zero, and Tuned-Lens-implied baselines. The ridge fit (layer-averaged 0.602) clears the pre-registered Tuned-Lens gate (-0.691) at every layer. The in-panel note ties the raw level to the tautology accounting of Section 3.4 (Figures 4 and 5), which any reader of this figure should hold alongside it.

independently per depth destroys genuine structure and tautology together), which is exactly why the pressure battery exists.

3.2 Cheap diagnostic: ridge from the state, against a calibrated lens

The protocol's first, cheap question: is Δ_ℓ linearly predictable from v_ℓ, better than trivial baselines *and* better than what a calibrated per-layer lens already implies? For each layer pair we fit low-rank ridge $\Delta_\ell \approx W v_\ell$ (ranks 16–128) on the train split and score held-out R^2 against: zero-prediction ($R^2 = -1.54$), per-layer mean ($R^2 = 0$ by construction), the identity-transparency regression of $v_{\ell+1}$ on v_ℓ ($R^2 = 0.965$; reported for transparency, dominated by the residual connection, not a comparator), and the **Tuned-Lens-implied update** $\hat{\Delta}_\ell^{\mathrm{TL}} = \hat{r}_{\ell+1} - \hat{r}_\ell$ from the public lens translators (Belrose et al., 2023), the pre-registered gate baseline for proceeding to the expensive AR tests.

Layer-averaged held-out R^2 is **0.602** at the best rank (128; 0.921 full-rank), versus -0.691 for the Tuned-Lens-implied update (Figure 1). The gate passes with a wide margin — the cheap diagnostic separates cleanly from calibrated per-layer refinement — so the AR programme is justified. A shrinkage-CCA backstop (protocol Step 4) finds held-out leading canonical correlations ≈ 0.9997 at every layer, collapsing to ≈ 0 under the scrambled null; we flag immediately that this near-1 is largely the definitional $-v_\ell \subset \Delta_\ell$ term amplified by the high-variance directions, so CCA confirms shared subspace and exposes nothing ridge missed — the substantive claim rests on the ridge-vs-baselines comparison.

3.3 H1 proper: autoregression on Δ and Granger reach

We fit pooled AR(k) models

$$\hat{\Delta}_\ell \;=\; \mu_\ell + \sum_{j=1}^{k} W_j\, \Delta_{\ell-j}, \qquad k \in \{1, 2, 3, 5, 10, 15, 20\},$$

with per-depth intercepts μ_ℓ (so the kernel cannot absorb per-depth mean offsets $\bar{\Delta}_\ell$ — essential for the Phase-2 stationarity comparison) and low-rank W_j shared across depth; predictors are compressed to the top-256 Δ principal components for tractability while the target stays full-D. Held-out R^2 reaches **0.65** at $k = 20$, rank 128, and is ≥ 0.47 at every k in the sweep (Figure 2).

How far back does depth dependence reach? At every feasible (depth ℓ, lag j) cell we run a nested-F lag test (Wilks' Λ with Rao's F, on a top-32 PCA projection since $D \gg N$ makes

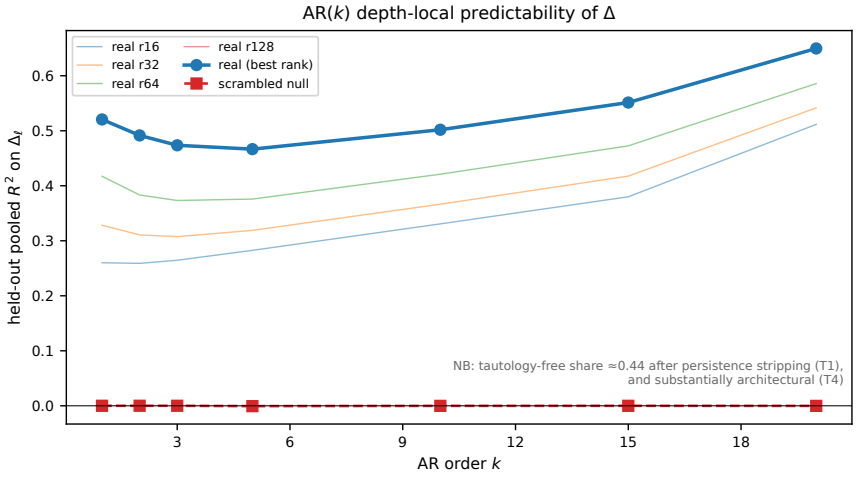


Figure 2: Pooled AR(k)-on-Δ held-out R^2 versus k (real vs the scrambled null) on Pythia-410m-deduped. Real ≥ 0.47 everywhere, peaking at 0.65; the null is indistinguishable from zero. The in-axes note carries the T1/T4/T5 correction: the raw level splits into tautological and genuine parts (Figures 4 and 5), and the genuine part is substantially architectural.

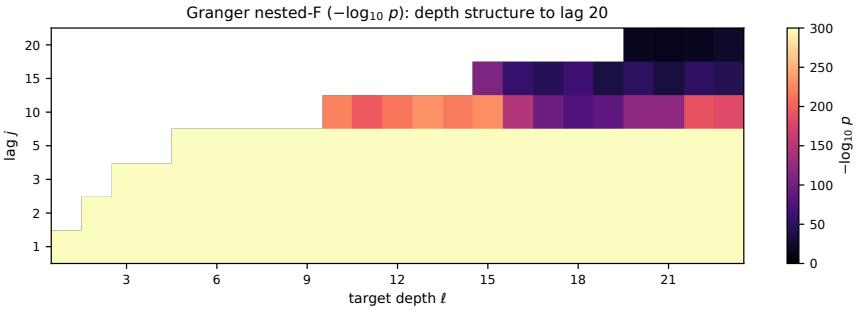


Figure 3: Granger-style nested-F significance ($-\log_{10} p$) per (depth, lag) cell. All 112 feasible cells are Bonferroni-significant out to lag 20; the scrambled null yields 0/112 (not shown). Statistically the depth dependence is long-range — reconstruction-wise it is first-order (Section 4.2), and Section 3.5 localises the long reach to the attention write.

full-D Wilks singular): does lag j add signal beyond lags $1, \ldots, j-1$ at depth ℓ? All **112/112** feasible cells are Bonferroni-significant ($\alpha = 4.5 \times 10^{-4}$ per cell), out to the longest lag tested, 20 (Figure 3). Section C verifies the F thresholds are calibrated with a Freedman–Lane permutation check (12/12 spot cells at the permutation floor; parametric and permutation tail quantiles agree within 1%).

Null 1 — scrambled activations (pre-registered). Permuting the example axis independently at each depth preserves every per-layer marginal and destroys within-example depth structure. The AR fit collapses to $R^2 \approx 0.0000$ at every k and the Granger grid to **0/112** significant cells. The signal is example-specific cross-depth structure, not a marginal artefact.

Null 2 — context perturbation (pre-registered, the critical control). Each prompt is wrapped in a fixed six-token neutral preamble; the perturbed variant replaces the preamble's content words with random function words at identical absolute positions. On the 863/1000 sampled examples where the wrapped baseline still recovers the answer, the AR fit is unchanged (0.639 wrapped $\rightarrow$ 0.696 perturbed) and the Granger heatmaps correlate at **0.964** (112/112 $\rightarrow$ 109/112 significant). The cross-layer dynamics at the query token are a property of subject–relation processing — depth-intrinsic, not leaked context.

3.4 Pressure-testing H1: how much is tautology, and is the rest learned?

Gate 1 passed on the pre-registered criteria. Before building on it, we attacked the headline with five post-hoc controls (T1–T5), then an untrained-network control (T4) — the corrected reading below, not the raw 0.65, is the version of H1 the rest of the paper uses.

T5 — exact tautology floor. The held-out explained variance of the single headline AR fit splits exactly (orthogonally, Pythagorean) across the top-m persistent state subspace and its complement. At $m = 50$: **factual** $0.65 = 0.52$ **tautological** $+ 0.13$ **genuine**; the persistent subspace itself is near-trivially predictable ($R^2_{\parallel} \approx 0.90$–$0.95$), which *is* the $\pm v_\ell$ drift made explicit (Figure 5).

T1 — persistence-stripped refit. Projecting out the top-m principal directions of the *states* and refitting: removing the single top direction (32% of state variance) costs only $0.650 \to 0.637$; removing the top 50 (87% of state variance) still leaves $R^2 = 0.436$, roughly $4000\times$ the scramble floor (Figure 4). T1 exceeds T5's genuine share because it re-optimises the kernel on the complement; both are reported.

T2 — persistence-matched surrogate. A random walk with independent per-layer Gaussian steps matched to each Δ_ℓ's mean and (rank-256) covariance — all persistence, no cross-layer dependence — scores $R^2 = \mathbf{0.0000}$. State persistence alone reproduces *none* of the AR-on-Δ predictability (for AR on increments, the $\mathrm{Var}(v_\ell)$ contributions to $\mathrm{Cov}(\Delta_\ell, \Delta_{\ell-1})$ cancel in expectation under independent steps), so the scramble floor was already the honest floor.

T3 — lag≥2 Granger, tautology-free by construction. The shared $\mp v_\ell$ term couples only lag 1 ($\Delta_{\ell-2}$ shares no state term with Δ_ℓ). Restricting the grid to lags ≥ 2: **89/89** cells significant (real) versus **0/89** (null), identically on factual and generic stores. Depth dependence beyond the mechanical overlap is total.

T4 — untrained-network control (deflationary). Running the identical battery on a *randomly initialised* Pythia-410m-deduped (same prompts, same pipeline): the persistence-stripped AR R^2 is **0.503 (untrained) vs 0.436 (trained)** on factual prompts (0.216 vs 0.290 on generic text), and the untrained network passes every tautology-free control — scramble ≈ 0, surrogate ≈ 0, lag≥2 Granger 89/89 — exactly like the trained one. The pre-registered inference rule ("trained $\gg$ untrained $\Rightarrow$ learned") returns the other branch: **the tautology-free AR-on-Δ structure is substantially architectural** — a property of the residual stack with per-layer nonlinear maps and LayerNorm, not of the learned weights. What training *does* change is the persistent/drift component (raw R^2 0.54→0.65 factual, 0.21→0.935 generic) — the part T1/T5 already attribute to the tautology. This replicates on GPT-2-small (trained 0.439 vs untrained 0.420 after stripping; learned share ∼5%), making the architectural reading a two-family statement.

3.5 A tautology-free localisation: attention vs MLP writes

Splitting each update into its two additive sublayer writes, $\Delta_\ell = \mathrm{attn}_\ell + \mathrm{mlp}_\ell$ (exact for Pythia's parallel blocks; additivity holds to 7×10^{-8} relative error), gives streams that carry *no* $\pm v_\ell$ term at all. Both independently reproduce H1 (AR real−null: attention **0.542**, MLP **0.653**; Granger 112/112 vs 0/112 both) — confirming the structure is not the persistence term — and they dissociate cleanly: the **MLP write is strictly first-order at every depth** ($k^* = 1$) while the **attention write carries the long-range depth memory** (k^* median 2, rising to 10

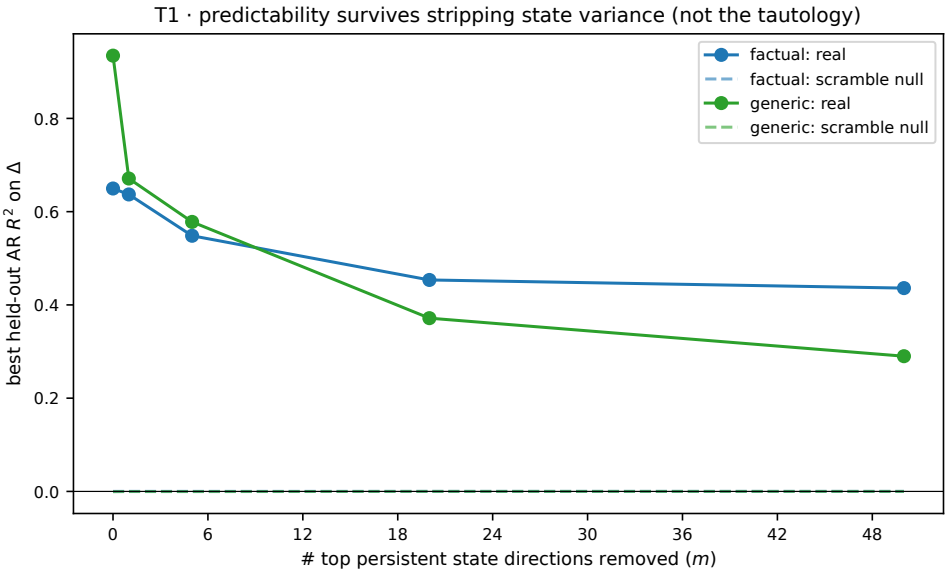


Figure 4: T1 persistence-stripped AR R^2 as the top-m state directions are removed, factual vs generic stores. The factual headline decays gradually and saturates at 0.44 $\gg$ null; the generic store — whose top direction alone is 96% of state variance — decays steeply to 0.29: the excess of the raw generic level (0.935) over factual (0.65) was persistence, the surviving structure is comparable.

late), with MLP contributing 78% of update energy. The residual stream's statistically-deep-but-reconstruction-shallow tension (Figure 3 vs Section 4.2) is therefore the energy-dominant memoryless MLP channel superposed on a small deep-reaching attention channel.

3.6 Robustness and the Gate-1 verdict

An object-disjoint split (grouping by answer object, since objects like "French" recur across up to 163 prompts while subjects are nearly unique) leaves the result intact: AR $R^2 = 0.561$, Granger 112/112, null ≈ 0. H1 also replicates on all six models (Section 4.4) and on length-matched generic Pile text (Section 4.5).

Gate-1 verdict (pre-registered routing): H1 supported; proceed to Phase 2 — with the corrected interpretation carried forward: genuine, above-null, depth-intrinsic cross-layer structure exists in the updates (T1–T3), but it is substantially a property of the architecture rather than of training (T4), and raw AR-on-Δ levels must never be quoted without the tautology split (T5).

4 Phase 2 — what kind of structure? Stationarity, order, and scope

4.1 H2 fails: the kernel is pervasively depth-dependent

H2 asks whether the AR kernel is the *same* kernel at every depth. The pre-registered test partitions depth into thirds (early ℓ 0–7, middle 8–15, late 16–23), fits one band-shared AR(k) kernel per band (per-depth intercepts μ_ℓ absorb mean offsets, so only transition structure is compared; $k = 5$, the plan's early-band feasibility cap $\lfloor L/3 \rfloor - 3$; working rank 256 after the plan's r-vs-$2r$ spot-check promoted 128), and compares the kernels four ways.

Every comparison rejects stationarity (Figure 6): *cross-band prediction* — within-band held-out R^2 is positive (0.74 / 0.58 / 0.67) but **every** off-diagonal is **negative** (e.g. early→middle −1.00, late→middle −1.89): a kernel from one band predicts another band's updates *worse*

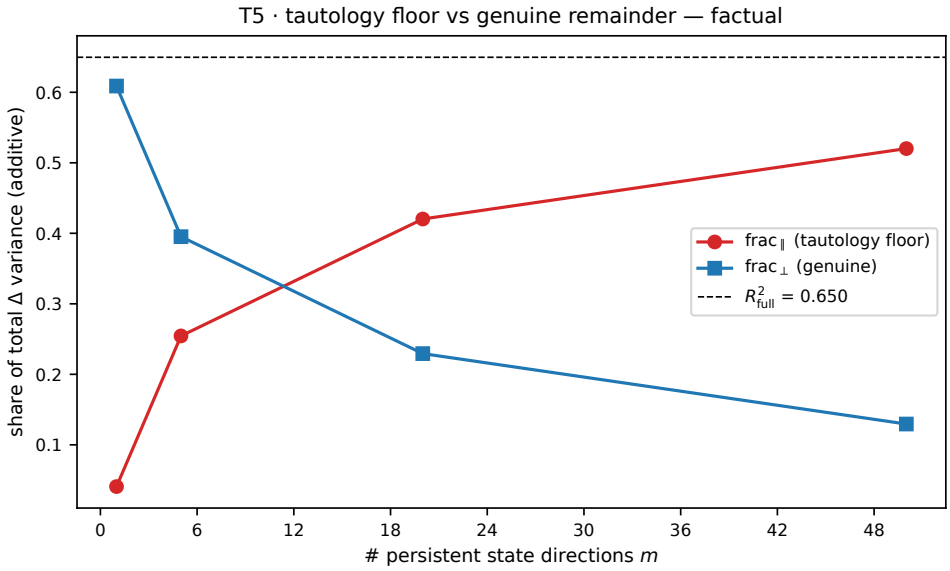


Figure 5: T5 exact decomposition of the single headline AR fit into tautological ($R^2_{\parallel}$, persistent-subspace) and genuine ($R^2_{\perp}$) shares as a function of the subspace size m. Factual: $0.65 = 0.52 + 0.13$ at m=50. Generic: $0.935 \approx 0.94 + 0$ — the generic level is persistence, not computation.

than that band's own mean. This holds at every truncation rank from 32 to 256 (worst off-diagonal −0.32 even at rank 32). *Frobenius distance* between band kernels is 1.39–1.42 $\approx \sqrt{2}$, the orthogonal-of-equal-norm ceiling; *principal angles* between kernel output subspaces average 57–62°; and a *bootstrap* within bands puts the between-band distance **80–104 standard deviations** above within-band sampling spread. Crucially there is *no clean phase boundary*: the three pairwise distances are equal to within 3%, so adjacent bands differ as much as distant ones — the depth dependence is **pervasive drift**, not segmentation.

Three caveats travel with this result: the cap makes it a short-horizon test (long-horizon stationarity is rechecked, and also fails, on the fitted generative model); AR-on-Δ stationarity is a proxy for H2's generative (MA) form — but a proxy whose failure is decisive in the rejecting direction; and the $\pm v_\ell$ shared term biases the comparison *toward* stationarity, so rejection is the conservative outcome.

Gate-2 verdict (pre-registered decision matrix): H2 fails pervasively ⇒ Phase 3 fits Branch B — a state-space model with *free per-layer operators* (the smoothly-varying variant assumes exactly the smoothness the data reject), and — per the proposal's own framing — CLT-style unconstrained per-(source, target) parameterisation is *validated*: "shared kernel along depth" is the wrong abstraction for these dynamics. The parsimony claim originally attached to H2 is off; what survives is a reconstruction comparison at stated parameter budgets (Section 6.4).

4.2 Effective order: $k^* \approx 1$, everywhere we looked

Per-depth AR(k) sweeps define the effective depth k^* as the smallest k within 95% of the asymptotic R^2. On the factual store $\boldsymbol{k^* \approx 1}$ at every depth (median 1; the single exception, k^*=2 at ℓ=17, coincides with the answer-promotion write discussed in Section 6.7): one preceding update captures essentially all held-out predictable variance (per-depth best R^2 0.83–0.94), lags ≥ 2 add a median +0.018, and the scrambled null collapses the whole curve to 0 (Figure 7). The contrast with the Granger reach of 20 (Figure 3) is a headline in itself: **long-range depth structure is statistically real but reconstruction-negligible** — and Section 3.5 locates the two halves of that sentence in different sublayers (deep reach: attention; reconstruction mass: MLP). k^* inherits the lag-1 tautology caveat, but it is also exactly the model-order information

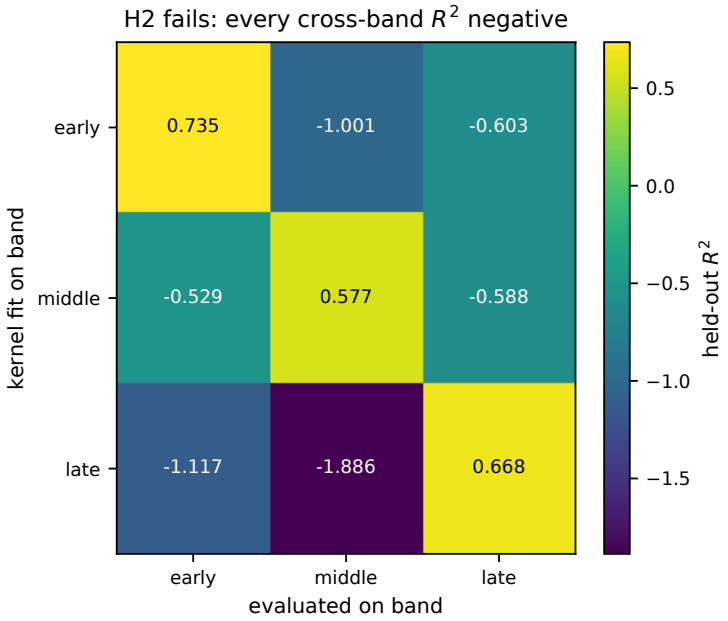


Figure 6: H2 stationarity test on Pythia-410m-deduped: held-out R^2 of band-shared AR(5) kernels evaluated within and across depth bands. Positive diagonal, every off-diagonal negative — a kernel from one depth band is worse than the target band's mean. Rank-robust; bands bootstrap-separated by 80–104 SD; no segment boundary.

Phase 3 needs, and the fitted model later supplies a mechanism for it (a contractive state operator, Section 6.8). Stratifying by relation class leaves $k^* = 1$ in every class and every depth band (class-invariant order); on generic text the median is $k^* = 2$ (band pattern 1/3/1) — stated explicitly since it differs from the factual 1, with per-depth generic fits being visibly noisier.

4.3 Scope: the kernel is query-position-specific and circuit-specific

Token position. Re-collecting trajectories at four positions per prompt (first subject token, last subject token, a mid-prompt non-subject token, final prompt token) and refitting AR(k^*=1) per position: within-position held-out R^2 is positive at every position (0.28–0.55), but **every cross-position R^2 is negative** (mean −0.36; kernels near-orthogonal, Frobenius $\approx \sqrt{2}$, angles 57–60°) — the same no-transfer signature as the depth bands, now across positions. The scrambled null collapses the diagonals, and the tautology (shared by diagonal and off-diagonal) cancels in the retention ratio. The Phase-1/2 characterisation is therefore scoped to *context-aggregated factual recall at the query position*; Phase 3 fits there, and every claim in this paper inherits that scope.

Relation class. The mirror-image test across the three relation classes (run with the fitted SSM of Section 5; reported here because it completes the scope picture): within-class held-out R^2 mean +0.835, **every cross-class transfer ≈ 0 or negative** (mean −0.191; Figure 8). The dynamics are finely tuned to each recall circuit — *circuit-specific kernels* — while the *order* $k^* = 1$ is class-invariant: same model order everywhere, different kernels per circuit.

4.4 Cross-model replication: six models, three families

The full H1/H2 pipeline (and, where public residual SAEs exist, the Phase-3 fit) was replicated on five further models; Pythia-70m additionally serves as the optimiser-debug lane and a depth point in Section 6.2. Table 1 shows the pattern is universal across the models tested: AR-on-Δ predictability far above its null with full Granger significance, every cross-band R^2 negative, and $k^* = 1$ at every band of every model.

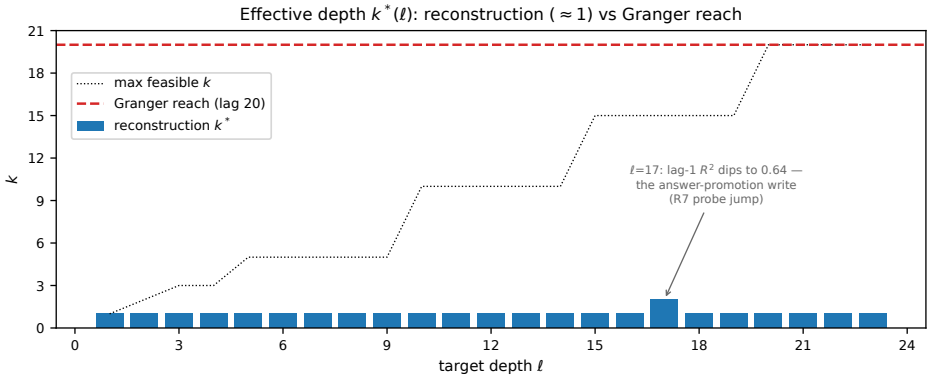


Figure 7: Per-depth effective order k^* on Pythia-410m-deduped (bars), with the per-depth lag-1 R^2 (line) and the Granger reach (20) for contrast. The one k^*=2 depth, ℓ=17, is the answer-promotion write: the R7 probe jump there (Figure 14) and the source-prefix sweep dips there (Figure 16).

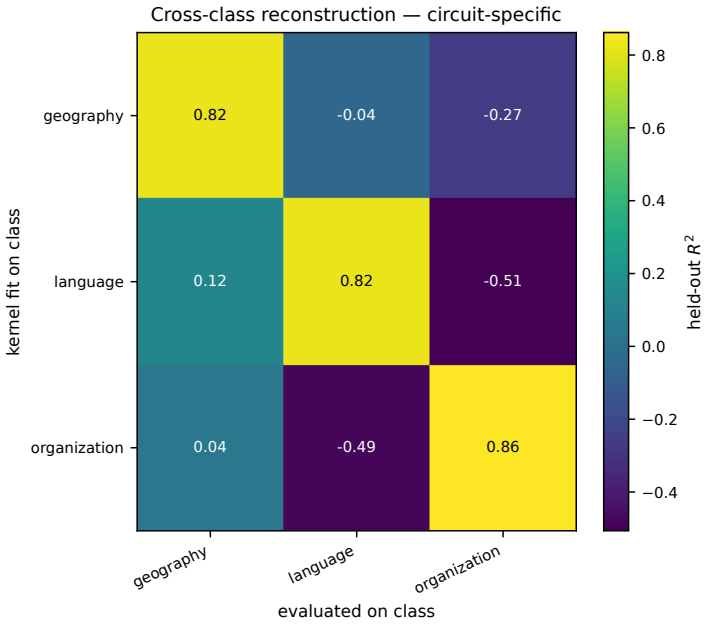


Figure 8: Cross-class transfer of the fitted decomposition (train class → eval class, held-out R^2). Strong diagonal, no transfer off it — the same signature as cross-band and cross-position: structured, but specific.

One anomaly is recorded rather than hidden: on GPT-2-small the cheap *low-rank* ridge diagnostic scores only 0.146 layer-averaged despite a full-rank 0.91 and AR 0.625 — GPT-2's $v \to \Delta$ map is far less low-rank than Pythia's. It does not affect any H1/H2 verdict.

4.5 Generality beyond factual recall

Re-running the identical pipelines on 1,555 generic Pile windows — length-matched to the factual prompt multiset (content lengths 4–24 tokens, median 9; one window per distinct document; this narrow-window construction is a stated scope condition) — replicates the *entire* characterisation: H1's binary claim holds (null ≈ 0, Granger 112/112 vs 0/112, lag$\geq$2 89/89), H2 fails even more emphatically (every cross-band R^2 negative, worst -144; bands separated by 39–188 bootstrap SD), and the effective order is low (k^* median 2). The raw generic AR level (0.935) is *higher* than factual (0.650), and Section 3.4 is the reason that must not be read as "more computation": T5 shows the generic level is ≈ 0.94 tautological + $\approx$0 genuine, and after persistence-stripping generic structure (0.29) is *weaker* than factual (0.44). The phenomenon is general to transformer

Table 1: Cross-model replication of H1/H2. AR R^2 is pooled held-out real−null; Granger counts are Bonferroni-significant cells (real vs scrambled null); H2 column reports the sign pattern of cross-band transfer. Datasets are per-model confidence-filtered CounterFact (sizes 767–2,540).

Model	L	AR R^2 (real−null)	Granger real / null	cross-band R^2	k^*
GPT-2-small	12	0.625	43/43 / 0	all < 0	1
Pythia-160m-deduped	12	0.894	43/43 / 0	all < 0	1
GPT-2-medium	24	0.513	112/112 / 0	all < 0	1
Qwen2-0.5B	24	0.462	112/112 / 0	all < 0	1
Pythia-410m-deduped (primary)	24	0.650	112/112 / 0	all < 0	1
GPT-2-large	36	0.311	196/196 / 0	all < 0	1

depth; the *kernels* remain position- and circuit-specific per Section 4.3.

5 The decomposition: a first-order, fully depth-indexed SSM over sparse sources

5.1 Form

Gate 2's decision matrix, applied to Section 4, fixes the architecture: pervasive (not smooth, not segmented) depth dependence rules out shared and interpolated kernels, and $k^* \approx 1$ makes the depth latent first-order. The generative model is a linear state-space model with free per-layer operators:

$$z_{\ell+1} = A_\ell z_\ell + B_\ell u_\ell, \qquad \hat{\Delta}_\ell = b_\ell + C_\ell z_\ell + D_\ell u_\ell, \tag{1}$$

where $u_\ell \in \mathbb{R}^{M_\ell}$ are sparse per-layer source codes, $z_\ell \in \mathbb{R}^{d_z}$ is the cross-layer state, b_ℓ is an unpenalised per-layer bias absorbing $\bar{\Delta}_\ell$, and every operator is rank-factored ($A_\ell = A_\ell^{(1)} A_\ell^{(2)\top}$, rank r; likewise B, C, D). The feedthrough $D_\ell u_\ell$ is the per-layer (no-propagation) path; the state path $C_\ell z_\ell$ carries whatever earlier layers contribute. The comparison that isolates the value of cross-layer propagation is therefore built in: the same model with the state ablated ($\hat{\Delta}_\ell = b_\ell + D_\ell u_\ell$, `use_state=False`) is a matched per-layer sparse dictionary sharing the optimiser, sources, sparsity sweep, and rank. Primary setting: $r = d_z = 32$.

5.2 Sources

Source codes initialise from public per-layer SAE latents of v_ℓ. On the primary model these are the MLSAE per-layer suite for Pythia-410m-deduped (Lawson et al., 2025) ($M = 65{,}536$, TopK k=32, all 24 layers); on the narrow query-position distribution only 497–4,433 latents per layer ever fire, so each layer is pruned to its alive set ($M_\ell^* \leq 4{,}433$) — standard dead-latent removal, exact for reconstruction, and legitimate here because the depth-indexed form permits per-layer widths. v_0 has no public embedding SAE, so u_0 is zero-initialised and free. Matched SAE state reconstruction averages $R^2 = 0.598$ across layers (two weak public checkpoints at v_{18}/v_{19}, 0.21/0.31, are shared by both comparison arms and cancel in every gap we report). Because the dictionary turned out to matter for identifiability (Section 6.5), we also train matched **L1** and **TopK** dictionaries on the same cached activations — identical architecture (per-token standardisation, tied decoder bias, unit-norm decoder columns), differing *only* in the sparsity mechanism (L1 coefficient 0.5 giving $L_0 \approx 36$, matched to TopK k=32) — including leakage-free variants fit on the train split only.

5.3 Optimisation and evaluation protocols

The loss is squared reconstruction error over depths and examples, plus an L1 penalty on all $u_\ell^{(i)}$ and a Frobenius penalty on the operators ($\lambda_{\text{reg}} = 10^{-4}$). Optimisation alternates

exactly as pre-registered: (i) a proximal-gradient LASSO pass over each example's stacked codes $(u_0, \ldots, u_{L-1})^{(i)}$ with operators frozen — FISTA with backtracking (Beck and Teboulle, 2009), validated against `sklearn` LASSO and closed-form least squares, keeping the exact zeros a penalised Adam would lose — and (ii) a full-batch Adam sweep over $\{A_\ell, B_\ell, C_\ell, D_\ell, b_\ell, z_0\}$ with codes frozen, differentiated through the unrolled SSM. Ten outer iterations; the method was debugged end-to-end on Pythia-70m before touching the primary lane.

Held-out numbers in this paper come from two protocols, and conflating them creates apparent contradictions, so we name them here and tabulate every operating point in Section B:

LASSO protocol (the Gate-3 protocol). Operators frozen after training; test codes re-inferred by the same LASSO against the test Δ. This is *reconstruction at matched sparsity* — the standard dictionary-learning evaluation — not out-of-sample forecasting; both comparison arms are treated identically.

Frozen-source protocol. Decoders fit by Adam on the *fixed* cached codes, no re-inference. Used for the CLT family (Section 6.4), where it isolates the decoder parameterisation; its absolute levels sit ∼0.08 below the LASSO protocol for every method.

5.4 Pre-registered sanity checks

All mandatory checks pass on the primary fit: no collapse to the bias solution (operator norms 252 vs bias norm 11; $L_0 \approx 0.009$); per-example reconstruction eyeball median $R^2 = 0.98$; top-20 features have mean relation-class purity 0.83 with human-legible themes (e.g. a "employed-by/manufactured-by → company" feature at layer 9, a "premiered-on → TV network" feature at layer 1, a "European place → Europe" feature at layer 22); dataset stability across disjoint halves top-100 Jaccard 0.89. Seed stability of decoder *directions* failed its pre-registered bar on the TopK source ($0.45 < 0.7$) — recorded at Gate 3 as a limitation, and subsequently traced to the dictionary type and closed, a story told in full in Section 6.5.

Gate-3 verdict: GO — the reconstruction clause is met with margin (next section), and the then-open seed-stability limitation was scoped to single-feature decoder identifiability, which no headline claim uses.

6 Evaluation

Unless stated otherwise, numbers are held-out R^2 on the fixed 311-example test split, and every headline gap carries a paired bootstrap 95% confidence interval over test examples (10,000 resamples; Section C). Model selection performed on the test split is limited to points where the contrast itself is the pre-registered quantity; the one CI where selection could matter (SSM vs CLT) is re-derived with validation-based selection in Section 6.4. Section B tabulates every operating point and which claim uses which.

6.1 Headline: cross-layer state beats a matched per-layer dictionary

Under the LASSO protocol, the SSM sits above its state-ablated twin at every sparsity level of the λ_{L1} sweep (Figure 9): held-out R^2 0.867–0.698 versus 0.816–0.653 across $\lambda_{L1} \in \{0.3, \ldots, 30\}$, an interpolated matched-L_0 gap of +0.037 **to** +0.066. At the canonical λ_{L1}=1 knee the gap is

+0.0489 [+0.0464, +0.0516] (MLSAE source),
+0.0483 [+0.0458, +0.0509] (trained-L1 source),

— dictionary-independent, seed-robust (every seed reaches $R^2 = 0.880$ train with the same gap), and it survives an object-disjoint split (+0.055/+0.061).

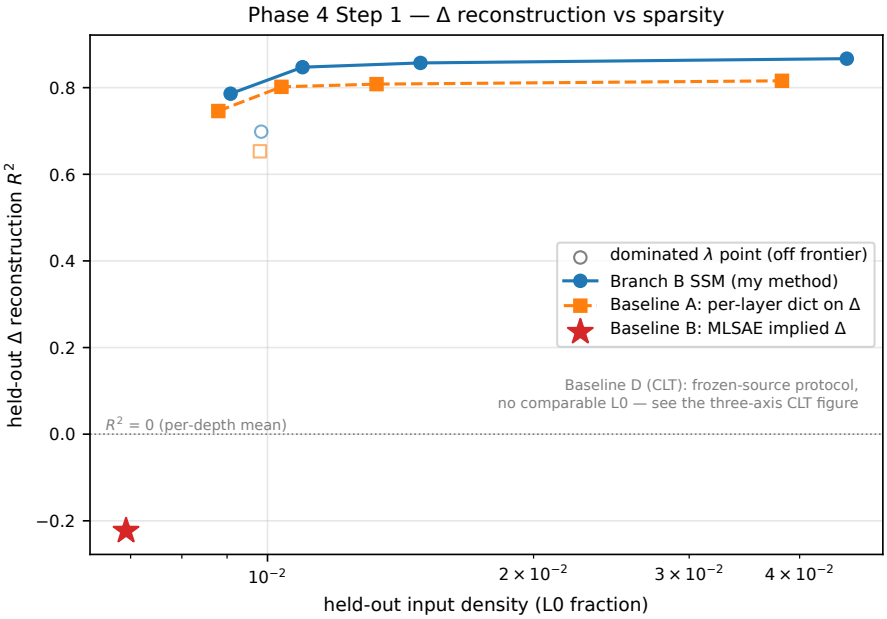


Figure 9: Sparsity–reconstruction Pareto on Pythia-410m-deduped (LASSO protocol): the SSM vs its state-ablated per-layer twin (Baseline A), with the naive differenced-SAE baseline (Baseline B, negative) for context (Section 6.3); the CLT lives in the frozen-source protocol with no comparable L_0 and gets its own figure (Figure 11). The SSM is above the matched baseline at every operating point.

Matched parameters, not just matched rank. The gap above compares arms at equal rank, giving the SSM roughly twice the baseline's parameters (8.36M vs 4.22M). Widening the baseline to rank 64 (8.41M) closes most of it: under the LASSO protocol the state-path edge survives at +0.0077 [+0.0053, +0.0102] — small, real, and conservative (the wider baseline runs slightly denser) — while under the frozen-source protocol the same comparison is a wash (±0.005, opposite signs at the 8.4M and 16.8M budgets). We therefore quote the headline as "+0.049 **rank-matched,** +0.008 **param-matched**" and never write "load-bearing" without the second number. What decoder width cannot replicate at any budget is the *structure* the state provides — Sections 6.7 and 6.8.

6.2 The gain grows with depth

Repeating the fit at every depth in the model zoo and reporting the *knee-gap* statistic (the raw same-λ gap at the $\lambda_{L1} = 1$ knee; the earlier max-over-λ statistic is upward-biased and seed-fragile because the baseline Pareto's tail saturates in L_0 — both statistics are archived) gives a monotone in-family depth series with tight 3-seed error bars:

$$\text{Pythia: } 6L \;\; +0.0076 \pm 0.0012 \;\rightarrow\; 12L \;\; +0.0130 \pm 0.0019 \;\rightarrow\; 24L \;\; +0.0504 \pm 0.0014,$$

with cross-family 12-layer points GPT-2-small $+0.0175 \pm 0.0025$ and Qwen2 $+0.0413 \pm 0.0025$ (Figure 10). Three caption-level caveats are part of this figure's claim: (i) the Pythia line mixes two TopK suites (EleutherAI-32k at 70m, MLSAE elsewhere); (ii) within a family, depth is confounded with width and parameter count (deconfounding via Pythia-1b/1.4b is recorded future work); (iii) the plotted Qwen point is the 12-layer *SAE-coverage prefix of a 24-layer model* — a dictionary-controlled contrast (our own L1 SAEs on the same rows at 24 vs 12 layers) gives +0.072 **at full depth vs** +0.043 **on the prefix**, so the public-suite point *understates* Qwen, and at true depth Qwen sits above Pythia-410m-deduped's +0.050.

How much of the gain is learned? Running the identical pipeline on a *randomly initialised* Pythia-410m-deduped (sources retrained on its own activations) yields a positive but smaller gap:

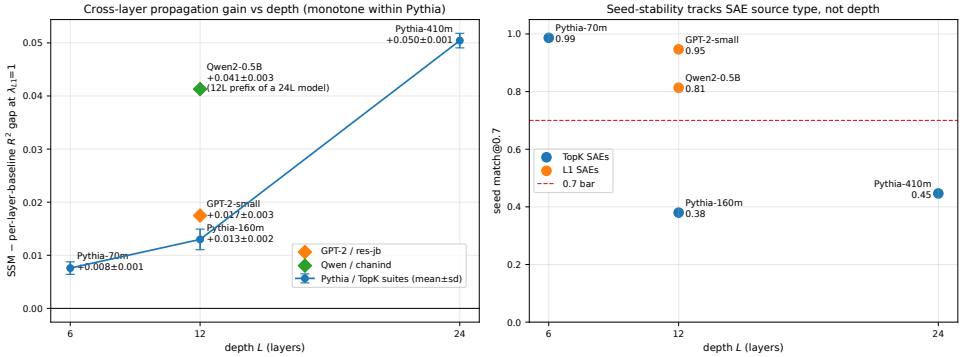


Figure 10: Cross-layer reconstruction gain (knee gap, SSM − matched per-layer baseline) versus depth: monotone within Pythia (3 seeds, mean±sd), replicated in sign in GPT-2 and Qwen2. Caveats in Section 6.2: mixed TopK suites on the Pythia line; in-family depth confounded with width; the Qwen point is a 12L SAE-coverage prefix of a 24L model (full-coverage value +0.072).

mean over the λ sweep +0.025 **untrained vs** +0.046 **trained** — the untrained network carries **53%** of the trained gain, and the learned share *grows with sparsity* (66% learned at λ_{L1}=10). The split replicates on GPT-2-small (untrained carries 44%). The dissociation with Section 3.4 is the point: AR-on-Δ *predictability* is almost entirely architectural (learned share ∼5% on GPT-2), while the SSM *reconstruction win* keeps a ≈half learned component on both families — the decomposition extracts something training actually adds, concentrated at the sparse operating points interpretability cares about.

6.3 Baselines that were not built for Δ, and it shows

The pre-registered indirect baselines both score *below zero* on update reconstruction: differencing the pretrained per-layer SAE's state reconstructions ($\hat{\Delta}_\ell = \hat{v}_{\ell+1} - \hat{v}_\ell$) gives $R^2 = \mathbf{-0.223}$ even though the same SAE reconstructs the *states* at +0.60 — the two reconstruction-error terms $\epsilon_{\ell+1} - \epsilon_\ell$ swamp the small update signal, the bias the protocol predicted, dramatic at 24 layers — and the Tuned-Lens-implied update gives $\mathbf{-0.309}$. The two Δ-targeted methods (SSM +0.867, matched per-layer dictionary +0.816) against the two indirect ones is the clearest statement of why the update needs to be modelled directly.

6.4 Against the unconstrained $O(L^2)$ parameterisation (CLT)

The primary conceptual comparison fits a *residual-write CLT* — a free low-rank decoder $W^{\ell' \to \ell}$ at every (source, target) pair, $\hat{\Delta}_\ell = \sum_{\ell' \le \ell} W^{\ell' \to \ell} u_{\ell'}$, the unconstrained family our SSM restricts — on the same frozen sources at the same per-decoder rank, under the frozen-source protocol. Scope qualifiers stated once and inherited by every sentence below: residual-write variant (not MLP-write), frozen shared sources (CLT does not get its own dictionary), per-pair rank capped at 32–64 (rank 128 exceeds the study GPU; Section E), $N = 1{,}244$ training examples, and no natively-trained CLT (limitation). The honest summary is three axes (Figure 11):

1. **Matched per-decoder rank: a small, real CLT edge.** At rank 32 the best-of-grid gap is SSM−CLT = −0.0142 [−0.0185, −0.0099]; re-selecting λ_{reg} on a train-internal validation split (never test) reproduces it at −0.0090 [−0.0136, −0.0045]. The edge exists at every depth tested (6L −0.009; 12L −0.050 GPT-2, −0.023 Pythia-160m; 24L −0.011) and is *not* monotone in depth — it tracks model and dictionary. (An earlier apparent tie was an artefact of a too-short regularisation grid; we report the corrected numbers.)

2. **Matched total parameters: the SSM Pareto-dominates up to 52M.** SSM rank-64 (16.9M parameters, R^2 0.8155) beats CLT rank-16 (26.2M, 0.794) *and* CLT rank-32 (52.4M,

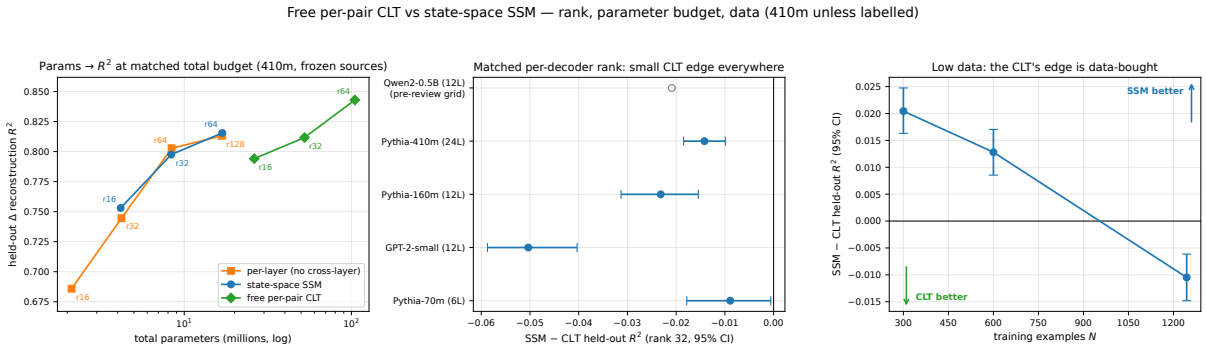


Figure 11: SSM vs residual-write CLT on frozen sources, three axes: (A) parameters → held-out R^2 at matched total budget (all decoder families near one frontier; per-layer rank-64/128 points included per Section 6.1); (B) per-model rank-matched gaps with paired CIs; (C) sample-efficiency sign flip (crossover $\approx 10^3$ examples).

0.804); the CLT needs 105M (rank 64, 0.843) to lead outright. Per-parameter, the first-order state is the better spend in this regime.

3. **Sample efficiency: the CLT edge is data-bought.** Subsampling the train set flips the sign: at N=300 the SSM leads by +0.0204 [+0.0163, +0.0248], at N=600 by +0.0128, at N=1,244 the CLT leads by 0.0105 — crossover at roughly 10^3 examples for 52M free parameters versus 8.4M shared ones.

Both cross-layer methods beat the no-propagation per-layer decoder at every depth tested (+0.022 to +0.061 across models; on the primary model SSM−per-layer +0.0518 [+0.0485, +0.0552] under validation-based selection). Read together with H2's failure, the conclusion is symmetric and, we think, useful in both directions: the free per-pair decoders are *validated* — they buy a small, significant, data-dependent margin, so nothing forces a shared kernel — and the first-order state-space compression captures most of the cross-layer reconstruction value at ∼16% of the parameters, dominating at matched budget and in low data.

6.5 Stability: the dictionary, not the model, controls identifiability

Gate 3 recorded one failed clause: top-100 decoder-direction stability across re-initialisations was 0.45 (match@0.7), below the pre-registered 0.7, while reconstruction was seed-identical — an equivalent-minima degeneracy, unmoved by 100× stronger operator regularisation and by shrinking d_z. Three findings close the story (Figure 12):

1. **It is the TopK sparsity mechanism.** Training matched L1 and TopK dictionaries on the *same* cached activations — identical architecture, matched L_0, differing only in the sparsity operator — and refitting: decoder match@0.7 is **1.00 (L1)** versus **0.08 (trained TopK)** versus 0.45 (pretrained MLSAE TopK), at the same reconstruction win (+0.041–+0.047). Not a width effect (the L1 dictionary is wider). Cross-model, L1 sources are stable on two further families (GPT-2-small 0.95, Qwen2 0.81) while TopK is unstable on two Pythia models (0.38, 0.45), with N ruled out by subsampling. Retraining the L1 dictionary on the train split only (no test leakage) changes nothing (match 1.00; gaps +0.037–+0.070).

2. **It is not a gauge artefact.** Gauge-invariant per-feature impulse responses (feedthrough plus downstream state path) are as unstable (0.29) as raw decoder columns (0.36), so the instability is genuine single-feature non-identifiability over redundant TopK atoms — expected for a distributed code where the fit only constrains realised combinations.

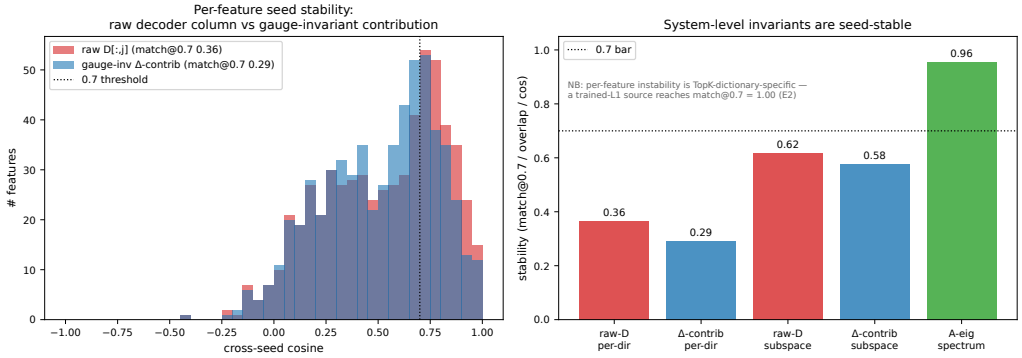


Figure 12: Per-feature vs system-level seed stability on the TopK (MLSAE) source. Left: cross-seed cosine of raw decoder columns (match@0.7 = 0.36) and of gauge-invariant per-feature Δ-contributions (0.29) — gauge-invariance does not rescue single-feature identifiability. Right: the system-level invariants the claims use are stable (operator spectrum, decoder subspace). The in-panel note carries the resolution: the per-feature instability is TopK-specific — a trained-L1 source on the same activations reaches match@0.7 = 1.00.

3. **Everything the claims use is stable.** Reconstruction (0.880 every seed), the state-operator eigenvalue spectrum (cosine 0.999 across seeds), the top-feature decoder *subspace* (∼0.6 overlap), and the feature *set* across data halves (Jaccard 0.89).

We flag the first finding as independently useful: pipelines that interpret *individual* SAE features while sourcing from TopK dictionaries are building on directions that need not survive a re-seed.

6.6 The causal existence proof: a clean, fully controlled null

The pre-registered critical test: find a seed-stable, interpretable feature, subtract its estimated contribution δ_ℓ from the residual stream during the real forward pass (re-solving the LASSO with the frozen decoder, so reconstruction error stays untouched), and require — all four pre-registered — a ≥ 0.20 mean top-1 probability drop on the feature's top-100 examples, preserved controls, a small matched-random-feature effect, and a symmetric ≥ 1.0-nat injection effect on 70% of silent examples.

The test fails, cleanly. Across the six strongest candidates (from a pool of 53 seed-stable features with purity 1.00 and legible themes), the largest ablation effect is +0.024 — ten times below threshold — and the injection test returns −0.02 nats; controls and magnitude-matched random features are ≈ 0, so the splice mechanism is clean. This is the pre-registered honest landing place, and we report it as such.

The diagnostics then locate *why*, with an escalating control sequence (Figure 13): amplifying one feature's δ_ℓ ten-fold drives the answer probability most of the way to zero, saturating at full removal by 30× (direction correct, natural magnitude small); ablating each example's top-k *co-firing* features crosses threshold at k=100 ($+0.237 \pm 0.035$; CI $[0.213, 0.262]$); a *count-matched* random group does not (+0.034) — but a *mass-matched* random group reproduces it (+0.244), and, decisively, a **norm-matched random direction** — even an isotropic one in $\mathbb{R}^D$ — reproduces it too (+0.224 isotropic, +0.204 contribution-shaped, crossing at the same k). Feature identity contributes a significant but ∼5% sliver ($+0.012$ $[+0.004, +0.021]$). Off-target specificity is weak and class-dependent (pooled on-target +2.89 vs off-target +2.45 nats; geography features answer-specific, organization features not). The group effect is therefore *magnitude-driven*: it demonstrates the query position's generic sensitivity to perturbations of that size, not a causally privileged code.

Swapping to the clean L1 dictionary strengthens rather than rescues the null: the candidate pool becomes 199/200 seed-stable, the maximum single-feature effect rises five-fold to +0.112

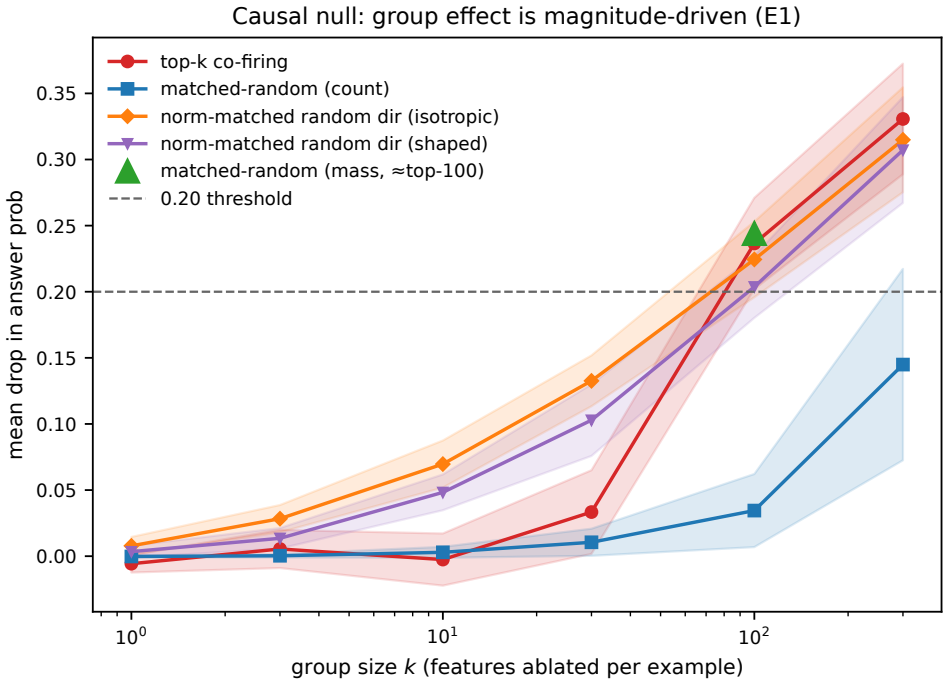


Figure 13: Group-ablation controls, pooled over 8 seed-stable candidates × 100 examples. Top-k co-firing ablation crosses the pre-registered 0.20 at k=100; count-matched random groups do not; mass-matched random groups and norm-matched random *directions* do — the effect tracks removed magnitude, not feature identity.

— a less redundant dictionary *concentrates* causal mass — and still no feature approaches 0.20. **Factual recall in this model is a distributed, mass-driven code with no single-feature bottleneck, independent of the dictionary used to read it.** No threshold was adjusted post hoc.

6.7 The state is functionally informative, not reconstruction-only

Does the 32-dimensional state z_ℓ carry task information, or is it only reconstruction machinery? We probe four arms at matched dimension d=32 with identical linear probes (hyperparameters selected on a train-internal validation carve-out): the state; PCA-32 of the current layer's code u_ℓ; PCA-32 of the concatenated prefix codes $u_{\leq \ell}$ (an information superset of what the state sees); and PCA-32 of the raw residual v_ℓ as a reference. Target: 205-way answer identity (majority 0.106; permuted-label null at majority).

At mid-depth the state wins against both code summaries on both dictionaries (Figure 14): state−code +0.060 [+0.037, +0.084] (MLSAE) / +0.139 [+0.105, +0.173] (L1); state−prefix +0.080/+0.173 at mid-depth and +0.110/+0.188 late; at ℓ=17 — the answer-promotion depth where k^* blips (Figure 7), the source-prefix sweep dips (Figure 16), and all probe arms jump — the state (0.759) beats even the raw-residual PCA (0.666). Peak state decoding reaches 0.775/0.801 top-1 from 32 dimensions. Since the SSM is linear, the state and the prefix-PCA are two 32-dimensional *linear images of the same prefix codes*: the compression trained only to reconstruct Δ retains answer-identity information that the variance-optimal compression discards. Scope, stated plainly: the late band washes out against a per-layer code summary, the raw stream remains the best readout late (−0.04 to −0.07), relation class saturates for every arm, and model confidence favours the code arms — the state's advantage is specific to *what* is being recalled, at the depths where recall is assembled.

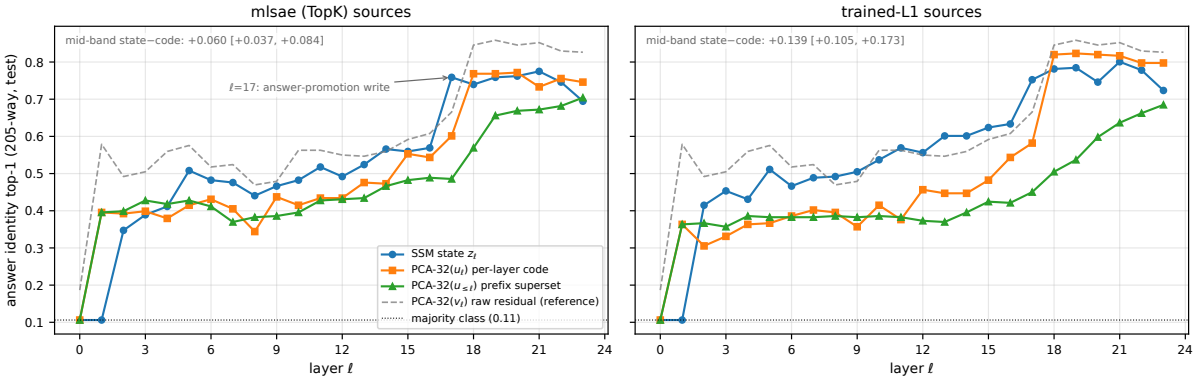


Figure 14: Answer-identity decoding (205-way top-1) per layer from four 32-dimensional summaries, both dictionaries. The Δ-reconstruction-trained state beats dimension-matched PCA of its own codes (and their prefix superset) at early/mid depth, and at ℓ=17 beats even raw-residual PCA; the raw stream remains best late.

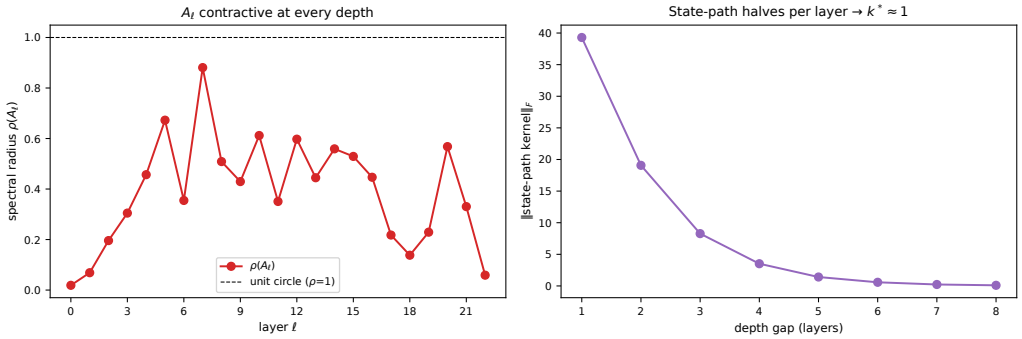


Figure 15: Spectrum of the learned state-transition operators A_ℓ: contractive at every depth (median $\rho = 0.43$), memory $\tau \approx 0.83$ layers, with the state-path kernel halving per layer of gap — the operator-level mechanism for $k^* \approx 1$. The spectrum is seed-stable (0.999) and dictionary-robust (0.993).

6.8 The learned propagation operator is contractive

The state-transition operators A_ℓ are the decomposition's structural readout, and their eigenvalues are gauge-invariant — a property of the fit's equivalence class, not of a seed. Three results (Figure 15): **(i)** the operator is **contractive at every layer**: spectral radius median 0.43, maximum 0.88, **0/23 layers** ≥ 1; implied spectral memory $\tau = -1/\ln \rho_{\text{geo}} \approx$ **0.83** layers. **(ii)** This is the *mechanism* behind $k^* \approx 1$: the state-path kernel norm halves per layer of gap ($1 \to 0.485 \to 0.211 \to 0.089$), so a source two layers back contributes half, three back a fifth — first-order reads capture essentially all state value. **(iii)** The spectrum is seed-stable (sorted-modulus cosine **0.999** across seeds) and dictionary-robust (L1 vs MLSAE **0.993**); honestly scoped, the per-layer ρ *profile* is only moderately shared across dictionaries (correlation 0.54) — the modulus distribution is pinned, the layer assignment less so. Depth-varying (H2 fails) + contractive + ∼1-layer memory is a compact, mechanistic summary of how these updates travel: *information written into the stream persists via the stream itself; the incremental dynamics forget almost immediately.*

6.9 Where the reconstruction lives: sources at every layer

Keeping sources only up to layer $\ell_{\max}$ and letting the SSM propagate them forward: held-out R^2 climbs monotonically from −0.44 (embedding-only; degenerate by construction since v_0 has no

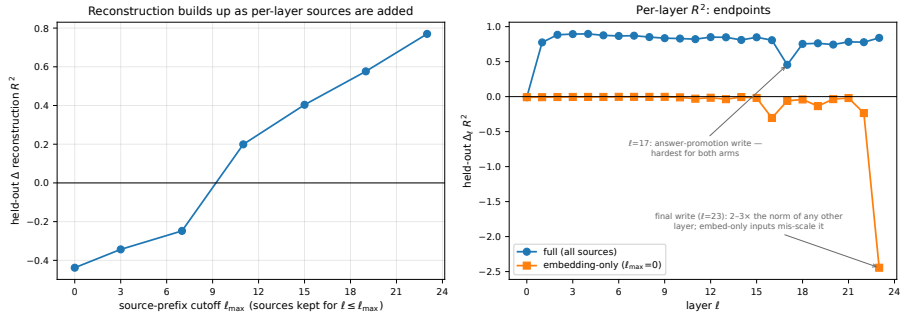


Figure 16: Source-prefix sweep: held-out R^2 as sources are admitted up to $\ell_{\max}$ (SSM free to propagate). Reconstruction only turns positive past mid-depth and is still climbing at the top of the stack — late-layer sources are load-bearing and non-propagable. The annotated interior dip at ℓ=17 is the answer-promotion write, the same depth where k^* blips (Figure 7) and the probe arms jump (Figure 14).

SAE) through +0.20 ($\ell_{\max}$=11) and +0.58 ($\ell_{\max}$=19) to +0.77 (full), still gaining +0.19 over the last four layers (Figure 16). Late-layer sources carry reconstruction value that propagation *cannot* substitute — the features are genuinely new at depth, matching the staged-recall picture of Geva et al. (2023) — which justifies the design's per-layer sources and completes a two-sided ablation ladder with the state ablation: removing propagation costs little (0.77 → 0.73 under this protocol); removing per-layer sources is catastrophic.

7 Limitations

Limitation 1 — feature novelty is structurally out of reach. The pre-registered novelty test (find a feature no baseline finds) reduces to a structural fact here: the SSM sources from the *same* SAE atoms as the per-layer baselines, so no feature can be novel by activation identity. The contribution is the cross-layer propagation structure over those atoms; no novelty fraction is computable or claimed.

Limitation 2 — cross-family coverage stops at 0.5B parameters. Every experiment ran on one 3 GB GPU (GTX 1050; Section E), which admits Pythia to 410m, GPT-2 to large, and Qwen2-0.5B, but not Gemma-2-2B or Llama-3.1-8B. The replication evidence is therefore three families at 6–36 layers; the two most architecturally distinct modern families are recorded future work, as is deconfounding depth from width within Pythia (1b/1.4b/2.8b).

Limitation 3 — three relation classes. The primary model confidently knows too few CounterFact facts for the protocol's original ≥1,000-per-class target, so the study is scoped to geography, language, and organization. No load-bearing claim depends on class breadth — the reconstruction and characterisation results are cross-model-replicated and the causal result is a controlled null — but cross-relation generality beyond these classes is untested, and the observed dissociation (organization features reconstruct class-specifically yet ablate non-specifically; geography features are specific on both) is reported on three classes without a breadth claim.

Scope of the characterisation. All kernels are fit at the *query position* of factual-recall prompts; cross-position transfer is negative (Section 4.3), so nothing here describes token positions in general. The generic-text replication uses short windows (median 9 content tokens) length-matched to the prompts. The attention/MLP split of Section 3.5 is exact only for parallel-block architectures like Pythia. Correctness-based probe targets are degenerate by construction (the dataset is filtered to model-correct facts), which is why the probe targets answer identity.

Methodological. The held-out protocol is reconstruction-at-matched-sparsity (codes re-inferred on test Δ with frozen operators), standard for dictionary learning but not out-of-sample forecasting; all arms are treated identically and Section B tabulates the operating points. Some hyperparameter selection touched the test split: AR best-(k, rank) is a max over symmetric real-vs-null evaluations (the contrast is the claim), and the CLT family's λ_{reg} was re-selected on a validation carve-out with the conclusion unchanged (Section 6.4); the Granger F approximation is empirically calibrated rather than assumed (Section C). The CLT comparison is the residual-write variant on frozen shared sources with per-pair rank ≤ 64; a natively-trained, jointly-sourced CLT could behave differently and remains the right off-box follow-up. Untrained-network controls use a single random initialisation each (their qualitative conclusions — never trained $\gg$ untrained for AR; untrained $\approx$ half for reconstruction — are far from the decision boundaries). Block-Toeplitz structure was always an approximation to test, not a derivation from the nonlinear architecture; H2's failure retires it.

Interpretive. H1's headline numbers are partly tautological and substantially architectural (Section 3.4); we consider the quantified version of that sentence a contribution, but it means raw AR-on-Δ levels from this or any similar pipeline should never be quoted as learned depth-computation. The causal null is a null for *this* feature-ablation methodology at *these* thresholds on *this* circuit family; it constrains single-feature stories of recall, not all causal uses of sparse decompositions.

8 Discussion

What kind of thing is depth structure? The composite picture is specific enough to be useful. Residual updates are predictable across depth, but the predictability decomposes into (i) a running-sum persistence term that is definitionally there, (ii) a genuine increment-level structure that a randomly initialised network already exhibits, and (iii) a learned component that only becomes visible in the *generative* fit — roughly half the SSM's reconstruction advantage, majority at sparse operating points. Meanwhile the dynamics are pervasively non-stationary in depth, first-order, and — per the fitted operator — contractive with a one-layer memory. A transformer, on this evidence, does not implement a depth-shared propagation rule; it implements a different small-memory rule at every layer, over features that persist via the additive stream itself rather than via the incremental dynamics. The Granger reach of 20 layers coexisting with $k^* \approx 1$ stops being a paradox once split by sublayer: a small, deep-reaching attention channel superposed on an energy-dominant, memoryless MLP channel.

Consequences for cross-layer interpretability methods. H2's failure is an independent validation of CLT's design decision to give every (source, target) pair free parameters (Ameisen et al., 2025) — our attempt to compress that family along depth was the pre-registered falsifier, and it failed in CLT's favour. The quantitative comparison then adds the budget-conditional half of the story: the free decoders buy a small, real, data-bought margin, while a first-order state captures most of the cross-layer value at a sixth of the parameters and generalises better below 10^3 examples. For practitioners the operational reading is: if you need cross-layer reconstruction under a parameter or data budget, a depth-indexed state is the better spend; if you need maximum fidelity and have data, free per-pair decoders are not redundant. Our TopK finding adds an orthogonal warning: single-feature decoder directions from TopK dictionaries were seed-unidentifiable in every configuration we tested, while matched L1 dictionaries were perfectly stable — any pipeline that names, steers, or ablates *individual* TopK features should check its features survive a re-seed.

Consequences for factual-recall stories. The causal null is not a failure to find structure; it is a measurement of what kind of structure recall has. Single features move the answer by at most a tenth of the pre-registered threshold; the effect that does clear threshold is reproduced by a norm-matched random direction; and a cleaner dictionary concentrates but does not localise the effect. This is a controlled, dictionary-independent statement that recall at the query position is a distributed, magnitude-dominated code — consistent with the drift in the editing literature away from single-component accounts (Hase et al., 2023), and a calibration point for ablation-based evaluation generally: group-ablation effects need magnitude-matched *direction* controls before they can be read as evidence about privileged feature sets.

Consequences for probing. The state-probe result cuts the other way: trained purely to reconstruct updates, the 32-dimensional state retains answer-identity information that variance-optimal compressions of the same codes discard. Reconstruction objectives, at least here, learn compressions aligned with what the circuit is computing — a modest but genuinely positive datum for the program of extracting functional summaries from decompositions, and one that pairs naturally with the contractive operator: a short-memory, information-dense running summary is exactly what a contractive state can be.

The EMG intuition, in retrospect. The convolutive-source-separation analogy (Holobar and Zazula, 2007) supplied H2, and H2 is false: transformer depth is not a homogeneous propagation medium. What survives from the analogy is the discipline it imported — model the increments, put sources at every position, and treat the mixing structure as an estimable, testable object. The estimated object just turned out to be depth-indexed, contractive, and circuit-specific rather than shared.

Future work. Three items are specified and costed in the project record: deconfounding the depth series from width (Pythia-1b/1.4b/2.8b with one dictionary type, three seeds); the two missing families (Gemma-2's sandwich normalisation, Llama-3.1-8B) which also enable the pre-registered $\geq$1,000-per-class relation test; and a natively-trained CLT. A fourth suggests itself from the results: whether the contractive-spectrum shape and the mid-depth functional-state window are universal across scale.

9 Conclusion

We pre-registered a hypothesis about how residual-stream updates travel through transformer depth, and the data answered with unusual precision: structured, yes — but the structure is substantially architectural where it is predictive, pervasively depth-indexed where we hoped it was shared, first-order and contractive where it is generative, distributed and magnitude-driven where we tested it causally, and genuinely learned exactly where the decomposition adds value over per-layer baselines. Both pre-registered hypotheses did real work: H1 by surviving controls designed to kill it, H2 by failing and thereby validating the unconstrained parameterisation the field already uses. Every gate decision, threshold, and fallback was written down in advance; every headline number carries a confidence interval; and the full pipeline — from cached activation tables to every figure in this paper — regenerates with one script. We offer the results, the nulls, and the protocol in equal measure.

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A Pre-registration and deviations

The protocol was written as four frozen phase documents (diagnostic, characterisation, decomposition, evaluation) plus a research statement, before any data were collected; a separate progress tracker was the only mutable record, so no pre-registered bar could be edited after the fact. The gates:

- **Gate 1** (H1 supported above null?) — *PASS*, all six pre-registered questions answered yes; routed to Phase 2.
- **Gate 2** (H2 stationarity?) — *FAIL (pervasive drift)*; routed, per the pre-registered decision matrix, to the state-space branch with free per-layer operators.
- **Gate 3** (reconstruction competitive at matched sparsity, and seed-stable features?) — *GO*: reconstruction clause met with margin; the seed-stability clause initially failed ($0.45 < 0.7$) and was recorded as a limitation per the plan's "reconstruction fine / seed unstable" branch, later closed by the dictionary-mechanism finding of Section 6.5.
- **Causal existence proof** — *NULL* at all four pre-registered thresholds; reported at the plan's designated honest landing place, with no threshold adjusted.

Deviations from the frozen protocol, exhaustively:

1. *Dataset thresholds* (plan-sanctioned tuning direction): CounterFact confidence threshold $0.3 \rightarrow 0.1$ and per-class floor $500 \rightarrow 300$, because the primary model knows only ∼2k facts; yield 1,555 examples / 3 classes instead of the aspirational 5–10k / ≥4 classes. Relation-class scope became Limitation 3 by an explicit recorded decision (option (b) of the scale-up fork) after the cross-model replications made class breadth non-load-bearing.

2. *Primary checkpoint*: Pythia-410m → Pythia-410m-*deduped* (the public Tuned Lens required for the pre-registered gate baseline exists only for deduped).

3. *Stationarity-test order cap* (in-plan): $k = 5$ from the plan's early-band feasibility rule; working rank promoted 128 → 256 by the plan's own r-vs-$2r$ spot-check.

4. *Cross-model targets*: Gemma-2-2B / Llama-3.1-8B do not fit the 3 GB study GPU; replaced by six models from three families that do (Limitation 2). The plan's Tuned-Lens gate was waived-and-reported on replication models without public lenses.

5. *Statistic repair* (post-hoc, both statistics archived): the depth-series gap statistic changed from max-over-λ matched-L_0 interpolation to the raw same-λ knee gap after per-seed forensics showed the former pairs against a saturated baseline tail; the knee statistic is the conservative one.

6. *Post-hoc analyses* beyond the protocol are labelled as such throughout (the T1–T5 pressure battery, untrained controls, sublayer split, parameter-matched baselines, the state probe, and the review-driven CI/validation-selection reruns).

B Operating points and evaluation protocols

All held-out R^2 values are against the per-depth-mean baseline on the same 311-example test split. The apparent spread of "the SSM's R^2" across sections is protocol, not inconsistency:

Table 2: Every SSM operating point quoted in this paper.

Value	Protocol	Setting	Used for
0.880	train fit	λ_{L1}=1, r=d_z=32	convergence/sanity
0.867	LASSO held-out	λ_{L1}=0.3	Pareto top end (Figure 9)
0.857	LASSO held-out	λ_{L1}=1 (knee)	headline gap +0.049
0.849	LASSO held-out	rank-64 *baseline*, λ_{L1}=1	param-matched gap +0.008
0.780–0.804	frozen-source	rank 32, best λ_{reg}	CLT family (Figure 11)
0.816	LASSO held-out	baseline (`use_state=False`)	Pareto baseline arm

The LASSO protocol re-infers test codes against test Δ with frozen operators (reconstruction at matched sparsity; identical for both arms of every gap). The frozen-source protocol fits decoders on fixed cached codes and isolates decoder parameterisation; its levels sit uniformly lower. Gaps are only ever computed within a protocol.

C Statistical details

Granger-style nested-F. At each (depth ℓ, lag j) cell we compare nested AR models (lags $1..j{-}1$ vs $1..j$, per-depth intercepts included) on a top-32 PCA projection of Δ (full-D Wilks is singular at $D \gg N$), using Wilks' Λ with Rao's F approximation. Bonferroni corrects across the 112 feasible cells ($\alpha = 4.5 \times 10^{-4}$ per cell); the feasibility count follows the plan's rule ($\ell \geq j$).

Permutation calibration. Because the rogue high-variance dimensions make errors heavy-tailed, the F thresholds were checked with a Freedman–Lane permutation scheme (Freedman and Lane, 1983): fit the reduced model, permute its residuals, recompute F (999 permutations), at 12 spot cells spanning mid/late depth and lags 1–20. All 12 observed F statistics sit at the permutation floor ($p \leq 10^{-3}$), and the permutation null's tail quantiles match the parametric $F(\text{df}_1, \text{df}_2)$ within 1% at the 95th/99th percentiles — the Bonferroni cut behaves as advertised under exchangeability. The scrambled-null grids (0/112 significant, both stores) are the complementary full-grid empirical calibration.

Paired bootstrap CIs. Every headline gap is a paired bootstrap over the 311 test examples (10k resamples) on per-example sums of squares, computed from saved per-example arrays — pure post-processing, no refitting. Seed-level uncertainty, where reported (Figure 10), is mean±sd over three full refits.

Selection. AR best-(k, rank) maximises held-out R^2 over 28 symmetric real-vs-null evaluations; the real−null contrast is the claim. The CLT family's λ_{reg} was originally selected on test (house convention, interior peaks); Section 6.4 re-derives the one consequential CI with validation-based selection, where every arm picks an interior λ^*=3 and the conclusion is unchanged (test selection had mildly flattered the CLT, not the SSM).

D Cross-model inventory

Table 3: Models, datasets, and source dictionaries. Confidence-filtered CounterFact sizes (70m: top-1,000 candidates by confidence — the optimiser-debug lane, excluded from the H1/H2 replication claim); SAE column gives the public suite used as frozen sources ("—" = diagnostics only). Seed-stability column is decoder match@0.7 across three fits.

Model	L	D	N	SAE source (type)	seed stab.
Pythia-70m-deduped	6	512	1,000	EleutherAI 32k (TopK k=16)	0.99
Pythia-160m-deduped	12	768	767	MLSAE x64-k32 (TopK)	0.38
Pythia-410m-deduped	24	1024	1,555	MLSAE x64-k32 (TopK) / trained L1	0.45 / 1.00
GPT-2-small	12	768	1,358	res-jb (L1)	0.95
GPT-2-medium	24	1024	1,795	—	—
GPT-2-large	36	1280	1,924	—	—
Qwen2-0.5B	24	896	2,540	chanind (L1, 12L prefix) / trained L1	0.81

The seed-stability column is the cross-model half of the Section 6.5 finding: L1 sources stable on two additional families, TopK unstable on two Pythia models, with dataset size ruled out by subsampling GPT-2 to the smallest N (match 0.95 unchanged).

E Reproducibility

All experiments ran on a single consumer GPU (NVIDIA GTX 1050, 3 GB, Pascal) with a pinned environment (Python 3.12; `torch` 2.5.1+cu124; `transformers` < 5; TransformerLens 2.x); the constraint is documented because it shaped the design (activation caching, alive-set compression, fp16 loading for the larger models, and the rank-128 CLT being infeasible). Every run logs its resolved config, git commit, seed, and library versions before doing any work. Numeric intermediates (CSV/JSON tables) are tracked in git; heavy artefacts (activation stores, checkpoints) are regenerable from scripts. A single entry point — `scripts/reproduce.sh` — checks the environment, runs the test suite (204 tests), asserts dataset sizes, and regenerates all 16 paper figures byte-identically from the tracked tables; `-toy-pipeline` additionally re-runs the full Pythia-70m lane end to end. Code, protocol documents, and the progress tracker are released with the paper under a permissive license.

Software: PyTorch (Paszke et al., 2019), TransformerLens (Nanda and Bloom, 2022), scikit-learn (Pedregosa et al., 2011), statsmodels (Seabold and Perktold, 2010), and the SAE suites cited in Section 5.2.